

Phase 4

Group 1

1. Project Overview

1.1 Project Summary

There are multiple movie platforms and evaluation systems (such as IMDb, Rotten Tomatoes, and major award institutions), each with different standards, audiences, and evaluation criteria. Because of these differences, it is often unclear how much a movie rating actually reflects its overall quality, popularity, or critical recognition. As a result, people may find it difficult to interpret movie ratings in isolation. By combining audience ratings, popularity measures, and award recognition, we aim to better understand what movie ratings represent and how different forms of recognition relate to each other.

1.2 Overarching Research Question

1. For movies released from 2016 to 2025, is there a relationship between the average IMDb rating and the average Rotten Tomatoes audience score?
2. For movies released from 2016 to 2025, do films that received Oscar recognition (i.e., were nominated or won at the Academy Awards) receive significantly higher audience ratings on IMDb and Rotten Tomatoes compared to films without any Oscar recognition?
3. For movies released from 2016 to 2025, is IMDb popularity (number of user votes) associated with the level of critic engagement on Rotten Tomatoes?

2. MongoDB Schema Design

In MongoDB, we use a primary collection called `merged_movies`. We embed data from IMDb, Rotten Tomatoes, and the Oscars into a single movie document, which reduces the need for join operations.

The top-level document represents a movie. Rating information from IMDb and Rotten Tomatoes is stored as embedded subdocuments in the form of string arrays. Oscar information is simplified into a boolean field named `oscar`, which indicates whether the movie has ever been nominated for an Oscar. We have copied the Rotten Tomatoes scores and Oscar award information into the documentation for each movie. This avoids the need for repeated

join operations across multiple collections, making it easier to run queries. In the final MongoDB design, we retained a collection named `merged_movies` for querying purposes. During data loading and transformation, we initially stored three separate collections: `movies`, `rotten`, and `oscars`, but the final design incorporates the relevant information directly into the movie documents rather than storing it separately.

```
{
  '-id': Objectid,
  'tconst': string
  'title': string
  'year': int
  'genres': string
  'imdb': {
    'rating': float
    'votes': int
  }
  'rotten': {
    'rating': {
      'audience': float
      'critics': float
    }
  }
  'oscar': boolean
}
```

Figure 1: MongoDB schema design for `merged_movies` collection

3. Research Questions

3.1 Research Question 1

Research Question

For movies released from 2016 to 2025, is there a relationship between the average IMDb rating and the average Rotten Tomatoes audience score?

MongoDB Query

```
pipeline1 = [  
  {  
    "$match": {  
      "year": {"$gte": 2016, "$lte": 2025},  
      "imdb.rating": {"$exists": True},  
      "rotten.ratings.audience": {"$exists": True}  
    }  
  },  
  {  
    "$project": {  
      "_id": 0,  
      "title": 1,  
      "year": 1,  
      "averageRating": "$imdb.rating",  
      "audience_score": "$rotten.ratings.audience"  
    }  
  }  
]
```

File Name and Line Number

File: phase4.py

Lines: 17 - 34

Explanation of the Query

MongoDB queries retrieve movies released between 2016 and 2025 and extract the IMDb rating and Rotten Tomatoes audience rating for each movie. It filters the dataset to ensure

that only movies with valid rating information from two sources are included. Query and output key attributes such as title, release year, IMDb average rating, and Rotten Tomatoes audience rating.

In Phase 3, the SQL version needs to connect the Movies and RottenTomatoes tables based on the title and release year. In contrast, the MongoDB implementation uses embedded documents to store this information in a single collection. Therefore, queries can directly access IMDb and Rotten Tomatoes data without performing explicit join operations.

The resulting dataset combines audience ratings of the same set of movies from two platforms, making it suitable for analyzing whether there is a relationship between IMDb ratings and Rotten Tomatoes audience ratings.

Result Produced by the Code

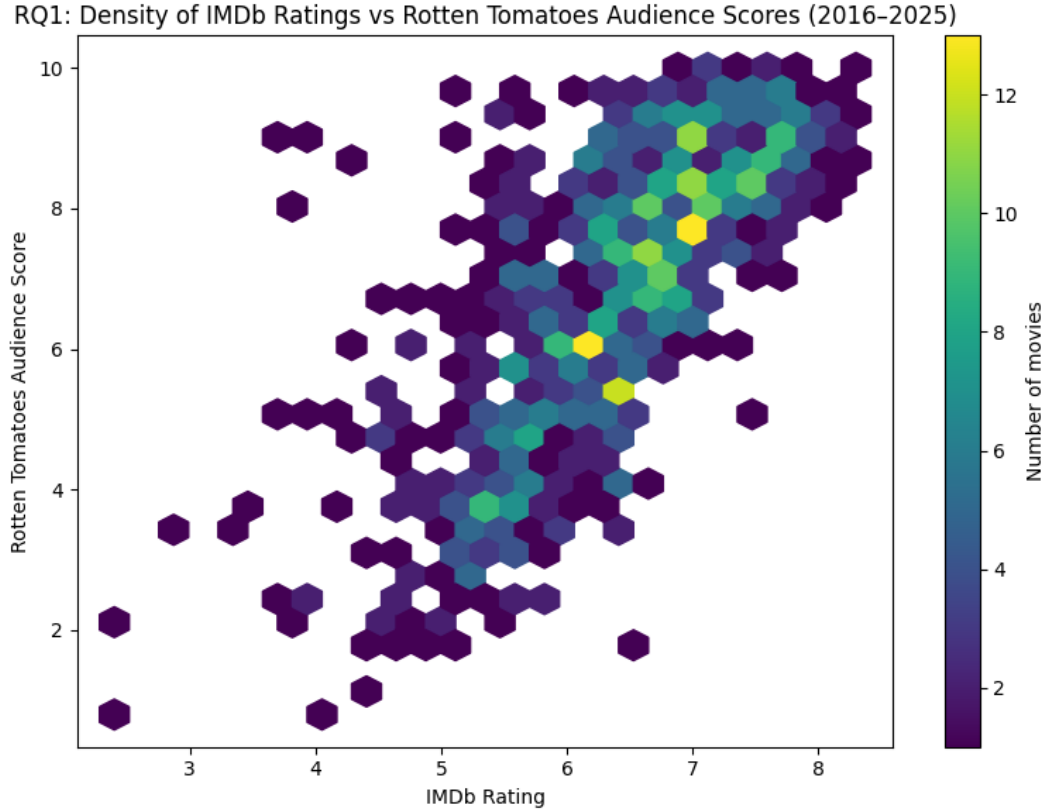


Figure 2: MongoDB Result for Research Question 1

Result Explanation

The results show a clear positive relationship between IMDb ratings and Rotten Tomatoes audience scores. As observed in both Phase 3 and Phase 4, movies with higher IMDb ratings

tend to also receive higher audience scores on Rotten Tomatoes.

3.2 Research Question 2

Research Question

For movies released from 2016 to 2025, do films that received Oscar recognition (i.e., were nominated or won at the Academy Awards) receive significantly higher audience ratings on IMDb and Rotten Tomatoes compared to films without any Oscar recognition?

MongoDB Query

```
pipeline2 = [  
  {  
    "$match": {  
      "year": {"$gte": 2016, "$lte": 2025},  
      "imdb.rating": {"$exists": True},  
      "imdb.votes": {"$exists": True},  
      "genres": {"$exists": True},  
      "oscar": {"$exists": True}  
    }  
  },  
  {  
    "$unwind": "$genres"  
  },  
  {  
    "$addFields": {  
      "popularity_bin": {  
        "$switch": {  
          "branches": [  
            {  
              "case": {"$lt": ["$imdb.votes", 10000]},  
              "then": "1K-10K"  
            },  
            {  
              "case": {"$lt": ["$imdb.votes", 100000]},  
              "then": "10K-100K"  
            }  
          ]  
        }  
      }  
    }  
  }  
]
```

```

        "case": {"$lt": ["$imdb.votes", 1000000]},
        "then": "100K-1M"
    }
    ],
    "default": "1M+"
}
},
"oscar_group": {
    "$cond": {
        "if": "$oscar",
        "then": "Oscar",
        "else": "No Oscar"
    }
}
}
},
{
    "$group": {
        "_id": {
            "genre": "$genres",
            "popularity_bin": "$popularity_bin",
            "oscar_group": "$oscar_group"
        },
        "avg_imdb_rating": {"$avg": "$imdb.rating"},
        "movie_count": {"$sum": 1}
    }
},
{
    "$project": {
        "_id": 0,
        "genre": "$_id.genre",
        "popularity_bin": "$_id.popularity_bin",
        "oscar_group": "$_id.oscar_group",
        "avg_imdb_rating": 1,
        "movie_count": 1
    }
},
{

```

```
        "$sort": {
            "genre": 1,
            "popularity_bin": 1,
            "oscar_group": 1
        }
    }
]
```

File Name and Line Number

File: phase4.py

Lines: 54 - 124

Explanation of the Query

The MongoDB query filtered movies released between 2016 and 2025, retaining only those with IMDb ratings. It first filters out valid documents, then uses `$unwind` to split the genre array so that each genre can be analyzed separately. Next, the query creates “Popularity” categories and “Oscar” tags, and calculates the average IMDb rating and number of films for each group. The final results illustrate how ratings vary across different genres, popularity levels, and Oscar award statuses.

Compared to SQL, the MongoDB version is more convenient because the data is embedded within a single collection. SQL requires join queries across different tables, whereas MongoDB can leverage the `$unwind` operator and aggregation stages to handle nested and array fields more naturally.

Result Produced by the Code

Result Explanation

The results indicate that movies with Oscar recognition generally receive higher IMDb ratings compared to those without Oscar recognition.

In Phase 3, the scatter plot (Figure 3) shows that Oscar-nominated movies are more concentrated in the higher rating range. This suggests a positive association between award recognition and audience ratings.

The MongoDB results in Phase 4 further confirm this pattern. Across different popularity bins and genres, movies with Oscar recognition consistently show higher average IMDb ratings than non-Oscar movies.

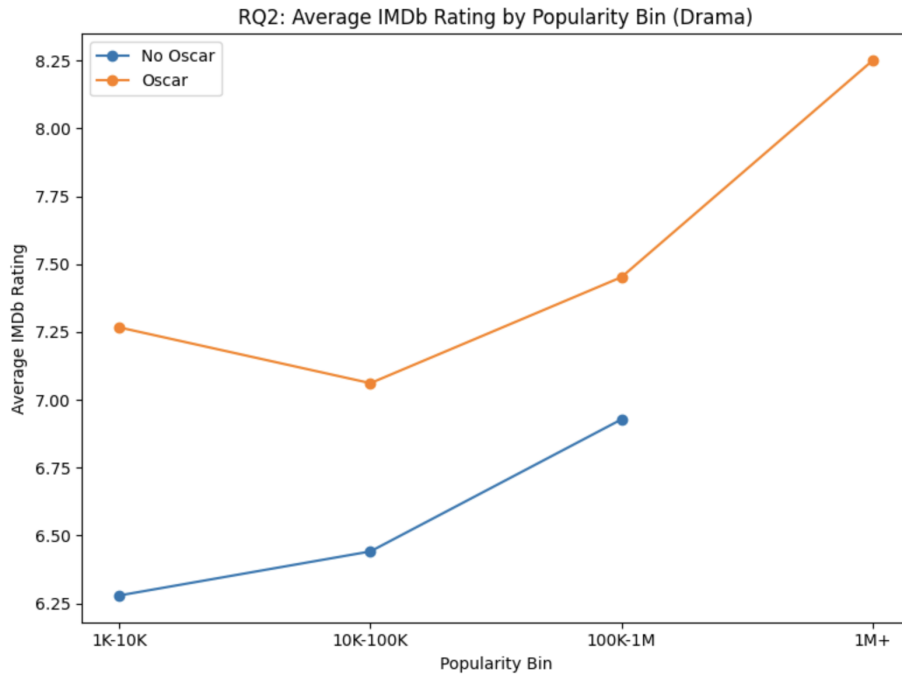


Figure 3: MongoDB Result for Research Question 2 (Part 1)

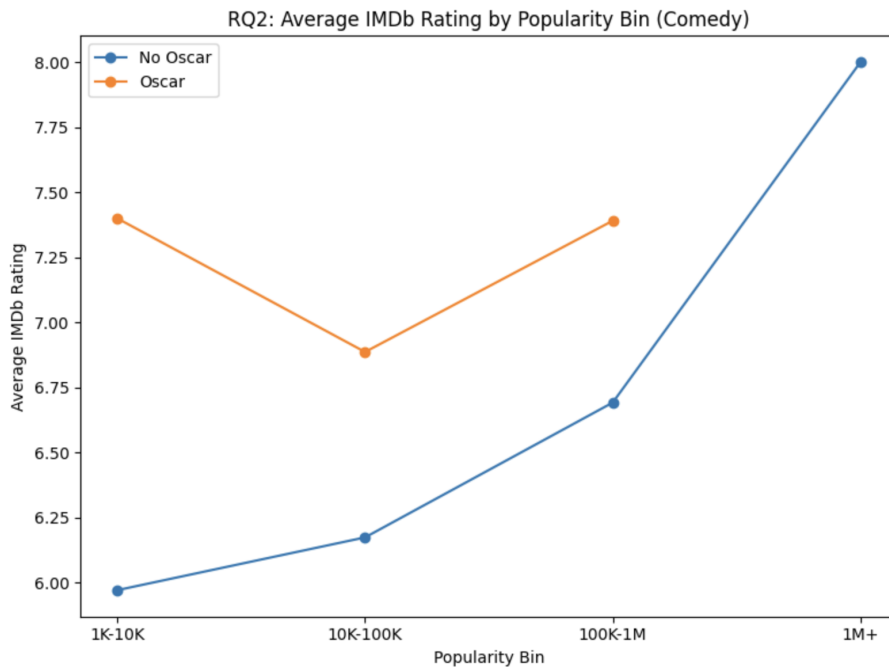


Figure 4: MongoDB Result for Research Question 2 (Part 2)

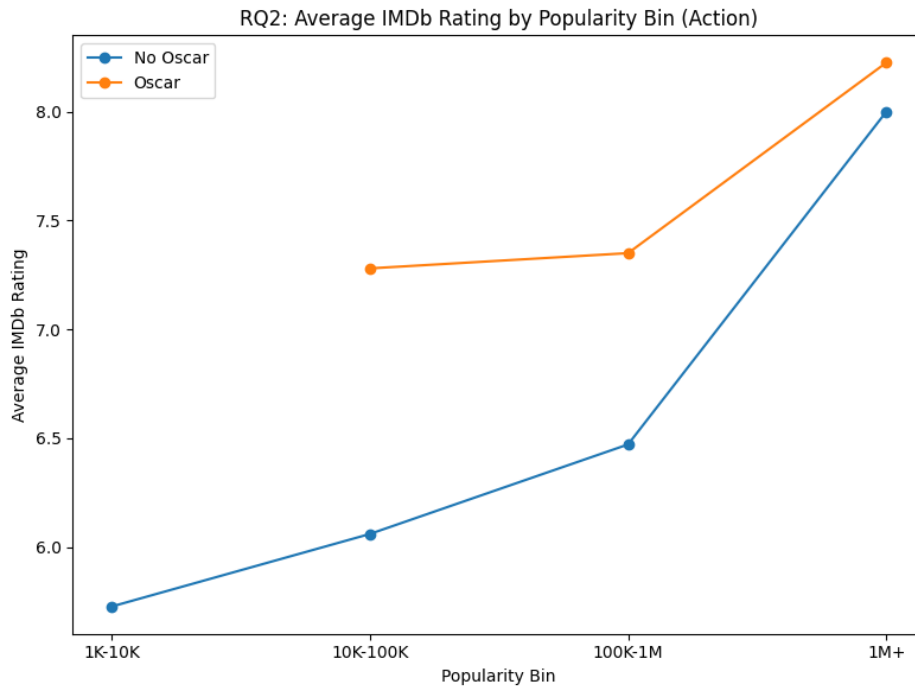


Figure 5: MongoDB Result for Research Question 2 (Part 3)

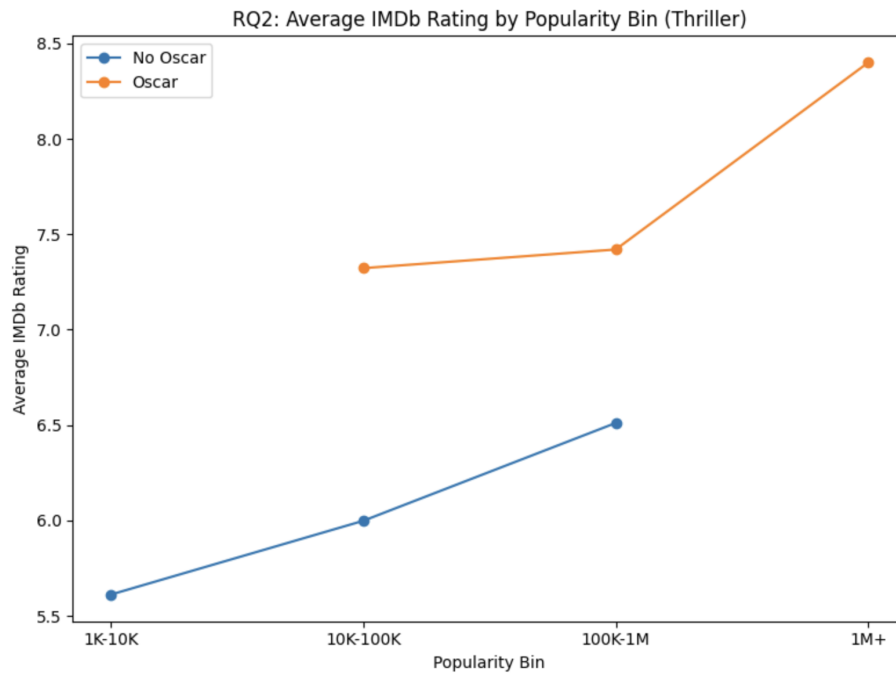


Figure 6: MongoDB Result for Research Question 2 (Part 4)

This relationship remains consistent after grouping by popularity levels, indicating that Oscar recognition is associated with higher perceived movie quality.

3.3 Research Question 3

Research Question

For movies released from 2016 to 2025, is IMDb popularity (number of user votes) associated with the level of critic engagement on Rotten Tomatoes?

MongoDB Query

```
pipeline3 = [  
    {  
        "$match": {  
            "year": {"$gte": 2016, "$lte": 2025},  
            "imdb.votes": {"$exists": True},  
            "rotten.ratings.critics": {"$exists": True}  
        }  
    },  
    {  
        "$project": {  
            "_id": 0,  
            "title": 1,  
            "year": 1,  
            "numVotes": "$imdb.votes",  
            "critics_score": "$rotten.ratings.critics"  
        }  
    }  
]
```

File Name and Line Number

File: phase4.py

Lines: 172 - 189

Explanation of the Query

This MongoDB aggregation pipeline filters documents based on their release years and ensures that both IMDb popularity (vote count) and Rotten Tomatoes critic ratings are avail-

able, thereby extracting movie-level observations of films released between 2016 and 2025. Then project the relevant fields, including the title, year, IMDb votes (numVotes) and Rotten Tomatoes score (critics_score).

Compared with the SQL version in stage 3, which requires adding the Movies and RottenTomatoes tables to the movie name and release year. MongoDB implements direct access to nested fields (imdb. votes and rotten.ratings. critics). In a collection of critics. This eliminates the need for join operations and simplifies the query structure. By placing numVotes and critics_score in the same result set for each film, the output provides the necessary pairwise variables to assess whether IMDb popularity is related to critic engagement. This makes the dataset suitable for correlation analysis and visualization.

Result Produced by the Code

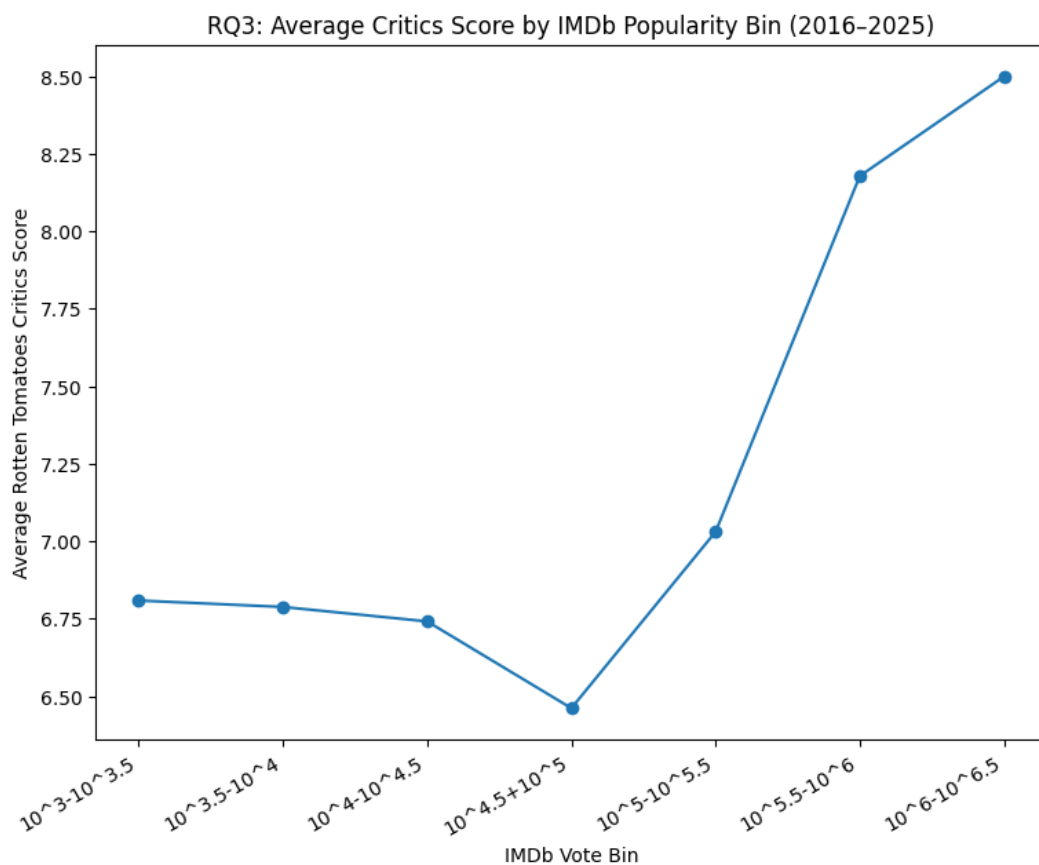


Figure 7: MongoDB Result for Research Question 3

Result Explanation

The relationship between IMDb popularity (number of votes) and Rotten Tomatoes critic scores appears to be weak and not strongly linear.

The results suggest a weak positive relationship between IMDb popularity (number of votes) and Rotten Tomatoes critic scores.

As shown in Phase 3, although movies with higher popularity tend to have slightly higher critic scores, the data points are widely dispersed and the trend is not strong.

The MongoDB results in Phase 4 confirm the same pattern. Therefore, IMDb popularity is not a strong predictor of critic engagement.